

CM3 - C LANGUAGE

Representations, programs, and builds

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COURSE SHAPE

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THE RULE FOR THIS DECK

Horizontal movement continues the lecture path.

Vertical movement opens details, examples, or exercises on the current topic.

In RevealJS, use right/left for the main path and down/up for deeper material.

TODAY AS A C COURSE

- C programs are data plus representation choices.
- Source files are translated, linked, and executed.
- Tools matter: compiler warnings, **make**, debugger, memory checker.
- The deck itself is a testbed for teaching code well.

WHAT THIS VERSION DEMONSTRATES

Feature	RevealJS	Beamer PDF	Typst handout
Vertical detail slides	interactive	printed as pages	printed as sections
Progressive line highlights	stepped	static fallback	static fallback
Code annotations	hover/selectable	below code	below code

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Feature	RevealJS	Beamer PDF	Typst handout
Scrollable long code	live only	avoid for final	avoid for final
Executed C example	executed at render	executed at render	executed at render

SOURCES AND ADAPTATION

This version combines:

- your imported CM3 material on files, compilation, `make`, tools, and style;
- the Berkeley CS61C lecture you added, especially number representations, C vs Java, `printf`, and undefined behavior;
- Quarto/RevealJS features that are useful for teaching code.

The Berkeley material is used as inspiration for topic order and examples, not as a slide import.

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VALUES AND BITS

REPRESENTATION IS A CONTRACT

A bit pattern has no meaning by itself.

`0b11111111` can mean:

- unsigned integer: `255`
- signed 8-bit integer: `-1`
- character byte:
implementation dependent
- raw mask: all bits set

The program works only if writer and reader agree on the representation.

```
read_byte.c
#include <stdint.h>
#include <stdio.h>

int main(void) {
    uint8_t byte = 0xff;
    printf("%u\n", byte);
}
```

FIXED-WIDTH INTEGERS

When the size matters, say it.

```
types.c
1  #include <stdint.h>
2  #include <stdio.h>
3
4  int main(void) {
5      uint8_t byte = 0xff;
6      int32_t temp = -12;
7
8      printf("byte=%u temp=%d\n", byte, temp);
9      return 0;
10 }
```

UNSIGNED ARITHMETIC WRAPS

wrap.c

```
1 #include <stdint.h>
2 #include <stdio.h>
3
4 int main(void) {
5     uint8_t x = 255;
6     x = x + 1;
7     printf("%u\n", x);
8 }
```

The result is 0 because unsigned arithmetic is modulo 2^N .

► Quick check

► Answer

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SIGNED OVERFLOW IS DIFFERENT

overflow.c

```
#include <limits.h>
#include <stdio.h>

int main(void) {
    int x = INT_MAX;
    printf("%d\n", x + 1);
}
```

This is not the same story as unsigned wraparound. Signed overflow is undefined behavior in C.

Warning

For teaching: this is a good moment to separate “what my machine did today” from “what C promises”.

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FIRST PROGRAM

ANATOMY OF `main`

hello.c

```
#include <stdio.h>
```

①

```
int main(int argc, char *argv[]) {
```

②

```
    printf("hello, CM3\n");
```

③

```
    return 0;
```

④

```
}
```

argv IS JUST DATA FROM THE SHELL

session.sh

```
$ ./show_args 1 hi
argv[0] = ./show_args
argv[1] = 1
argv[2] = hi
```

show_args.c

```
1 #include <stdio.h>
2
3 int main(int argc, char *argv[]) {
4     for (int i = 0; i < argc; i++) {
5         printf("argv[%d] = %s\n", i,
6             )
7         return 0;
8     }
```

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printf: INTEGER FIRST

printf.c

```
int n = 42;
printf("value = %d\n", n);
```

%d asks **printf** to read the next argument as a signed decimal integer.

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printf: THEN FLOATING POINT

printf.c

```
double x = 42.0;
printf("value = %f\n", x);
```

%f asks **printf** to read the next argument as a floating-point value.

FORMAT STRINGS ARE CONTRACTS

wrong-format.c

```
1 #include <stdio.h>
2
3 int main(void) {
4     double x = 42.0;
5     printf("as int: %d\n", x);
6     printf("as double: %f\n", x);
7 }
```

The compiler may warn, but **printf** itself trusts the format string.

COMMON `printf` SPECIFIERS

Specifier	Meaning	Typical argument
<code>%d</code>	signed decimal integer	<code>int</code>
<code>%u</code>	unsigned decimal integer	<code>unsigned int</code>
<code>%x, %X</code>	hexadecimal integer	<code>unsigned int</code>
<code>%f</code>	floating point	<code>double</code>
<code>%s</code>	string	<code>char *</code>
<code>%p</code>	address	<code>void *</code>
<code>%%</code>	literal percent sign	none

COMPILATION

C IS COMPILED FOR A TARGET MACHINE

C: Java/Python comparison:

- source is translated before execution;
- output depends on architecture and system ABI;
- compiler warnings are part of the feedback loop.
- Java source becomes bytecode, then JIT/runtime work happens;
- Python usually compiles to bytecode at runtime;
- portability is paid for by a runtime.

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THE BUILD PIPELINE

build.sh

```
1 $ gcc -E hello.c -o hello.i
2 $ gcc -S hello.i -o hello.s
3 $ gcc -c hello.s -o hello.o
4 $ gcc hello.o -o hello
```

- Preprocess: expand directives like `#include`.
- Compile: turn C into assembly.
- Assemble: turn assembly into an object file.
- Link: combine objects and libraries into an executable.

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QUARTO CAN EXECUTE A C BUILD AT RENDER TIME

The generated RevealJS/Beamer/Typst output is static, but this cell compiles and runs C while the document is rendered.

```
$ gcc -Wall -Wextra hello.c -o hello
$ ./hello
hello from compiled C
```

STATIC TRACE IS STILL USEFUL

Use a plain **bash** code block when the goal is a readable transcript rather than live execution.

```
session.sh
$ gcc -Wall -Wextra hello.c -o hello
$ ./hello
hello from compiled C
```

This is deterministic, portable, and easy to print.

MULTI-FILE PROJECTS

A HEADER IS THE PUBLIC CONTRACT

counter.h

```
#ifndef COUNTER_H
#define COUNTER_H

extern int counter;
int next_value(void);

#endif
```

counter.c

```
#include "counter.h"

int counter = 0;

int next_value(void) {
    return ++counter;
}
```

INCLUDE DECLARATIONS, NOT DUPLICATE DEFINITIONS

main.c

```
1 #include "counter.h"
2 #include <stdio.h>
3
4 int main(void) {
5     printf("%d\n", next_value());
6     return 0;
7 }
```

The header lets each `.c` file compile independently while agreeing on shared declarations.

LONGER EXAMPLE WITH SCROLLING

This slide intentionally scrolls in RevealJS so we can discuss a complete small project without splitting context.

counter.h

```
#ifndef COUNTER_H
#define COUNTER_H

extern int counter;
int next_value(void);

#endif
```

counter.c

```
#include "counter.h"

int counter = 0;

int next_value(void) {
    return ++counter;
}
```

report.c

```
#include "counter.h"
#include <stdio.h>

void report_value(const char *label) {
    printf("%s: %d\n", label, counter);
}
```

main.c

```
#include "counter.h"

void report_value(const char *label);

int main(void) {
    report_value("initial");
    next_value();
    report_value("after one step");
    next_value();
    report_value("after two steps");
    return 0;
}
```

BUILD AUTOMATION

SEPARATE COMPILATION

build.sh

```
1 $ gcc -Wall -Wextra -c main.c -o main.o
2 $ gcc -Wall -Wextra -c counter.c -o counter.o
3 $ gcc -Wall -Wextra -c report.c -o report.o
4 $ gcc main.o counter.o report.o -o demo
5 $ ./demo
```

Object files let us rebuild only what changed.

A MAKEFILE RECORDS DEPENDENCIES

Makefile

```
1 CC = gcc
2 CFLAGS = -Wall -Wextra -g
3 OBJ = main.o counter.o report.o
4 EXEC = demo
5
6 all: $(EXEC)
7
8 $(EXEC): $(OBJ)
9 ____$(CC) $(OBJ) -o $$@
10
11 main.o: main.c counter.h
12 ____$(CC) $(CFLAGS) -c $< -o $$@
13
14 counter.o: counter.c counter.h
15 ____$(CC) $(CFLAGS) -c $< -o $$@
16
17 report.o: report.c counter.h
18 ____$(CC) $(CFLAGS) -c $< -o $$@
```


FULL MAKEFILE WITH TESTS

This is the kind of reference slide where vertical scrolling can be useful during live teaching.

Makefile

```
CC = gcc
CFLAGS = -std=c17 -Wall -Wextra -Wpedantic -g
CPPFLAGS = -Iinclude
LDFLAGS =

SRC_DIR = src
TEST_DIR = tests
BUILD_DIR = build

APP = $(BUILD_DIR)/counter-demo
TEST_APP = $(BUILD_DIR)/counter-tests

APP_SRC = $(SRC_DIR)/main.c \
          $(SRC_DIR)/counter.c \
          $(SRC_DIR)/report.c

TEST_SRC = $(TEST_DIR)/test_counter.c \
           $(SRC_DIR)/counter.c

APP_OBJ = $(APP_SRC:%.c=$(BUILD_DIR)/%.o)
TEST_OBJ = $(TEST_SRC:%.c=$(BUILD_DIR)/%.o)

.PHONY: all test clean distclean run

all: $(APP)

run: $(APP)
____./$(APP)

test: $(TEST_APP)
____./$(TEST_APP)

$(APP): $(APP_OBJ)
____$(CC) $(APP_OBJ) $(LDFLAGS) -o $@

$(TEST_APP): $(TEST_OBJ)
____$(CC) $(TEST_OBJ) $(LDFLAGS) -o $@

$(BUILD_DIR)/%.o: %.c include/counter.h
____mkdir -p $(dir $@)
____$(CC) $(CPPFLAGS) $(CFLAGS) -c $< -o $@
```

clean:

```
____rm -rf $(BUILD_DIR)/**/*.o
```

```
distclean:
```

```
____rm -rf $(BUILD_DIR)
```

THE TAB MATTERS

Makefile

```
clean:
```

```
____rm -f *.o demo
```



The command line under a target starts with a real tab character.



Tip

For class, make the invisible tab visible once, then stop talking about it and let **make** enforce the rule.

WHAT make CHANGES

session.sh

```
$ make
gcc -Wall -Wextra -g -c main.c -o main.o
gcc -Wall -Wextra -g -c counter.c -o counter.o
gcc -Wall -Wextra -g -c report.c -o report.o
gcc main.o counter.o report.o -o demo

$ touch report.c
$ make
gcc -Wall -Wextra -g -c report.c -o report.o
gcc main.o counter.o report.o -o demo
```

Only the affected object and the final executable are rebuilt.

UNDEFINED BEHAVIOR

QUIZ: UNINITIALIZED LOCAL VARIABLE

ub.c

```
1 #include <stdio.h>
2
3 int main(void) {
4     int x;
5     printf("%d\n", x);
6     return 0;
7 }
```

What should students answer?

- Not “0”.
- Not “a random number” as a language guarantee.
- The correct C answer is: the program has undefined behavior.

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UNDEFINED BEHAVIOR BREAKS REASONING

Undefined behavior can:

- appear stable on one machine;
- change after optimization;
- change when a debugger is attached;
- disappear when a print is added;
- survive tests and fail later.

The teaching rule: compile with warnings, remove undefined behavior first, then debug logic.

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USE BRACES

braces.c

```
1 if (ready)
2     start();
3
4 if (ready) {
5     start();
6 }
7 cleanup();
```

The second form makes scope visible and prevents future edits from changing control flow by accident.

C TRUTHINESS

truth.c

```
if (0) {
    puts("false");
}

if (42) {
    puts("true");
}

if ("hello") {
    puts("also true");
}
```

In C17, include `<stdbool.h>` if you want `bool`, `true`, and `false`.

TOOLS

WARNINGS ARE PART OF THE COURSE

session.sh

```
$ gcc -Wall -Wextra -Wpedantic -g ub.c -o ub
ub.c: In function 'main':
ub.c:5:5: warning: 'x' is used uninitialized [-Wuninitialized]
```

Do not treat warnings as decoration. In this course, students should read them from the top.

SCROLLABLE DIAGNOSTICS TRACE

This slide is intentionally too long for a fixed slide. The point is to practice reading from the first meaningful diagnostic.

```
diagnostics.sh
$ gcc -std=c17 -Wall -Wextra -Wpedantic -g src/main.c src/counter.c -o demo
src/main.c: In function 'main':
src/main.c:8:5: warning: implicit declaration of function 'report_value' [-Wimplicit-function-declaration]
    8 |     report_value("initial");
      |     ^~~~~~
src/main.c:10:17: warning: format '%d' expects argument of type 'int', but
argument 2 has type 'double' [-Wformat=]
   10 |     printf("%d\n", 3.14);
      |             ~^   ~~~~
      |             |     |
      |             int  double
      |             %f
src/main.c:12:9: warning: unused variable 'unused' [-Wunused-variable]
   12 |     int unused = 0;
      |     ^~~~~
src/counter.c: In function 'next_value':
src/counter.c:7:1: warning: control reaches end of non-void function [-Wreturn-type]
    7 | }
      | ^
/usr/bin/ld: /tmp/ccxT4y0F.o: in function 'main':
src/main.c:8: undefined reference to 'report_value'
/usr/bin/ld: src/main.c:9: undefined reference to 'report_value'
collect2: error: ld returned 1 exit status
```

Start at the first warning that explains a real source problem, then rebuild after each fix.

DEBUGGER WORKFLOW

`gdb-session.sh`

```
$ gcc -Wall -Wextra -g main.c counter.c report.c -o demo
$ gdb ./demo
(gdb) break main
(gdb) run
(gdb) next
(gdb) print counter
(gdb) backtrace
```

Good debugger slides are usually command traces plus a small code fragment, not screenshots.

MEMORY CHECKER WORKFLOW

`valgrind-session.sh`

```
$ valgrind --tool=memcheck ./demo
==12345== HEAP SUMMARY:
==12345==    in use at exit: 16 bytes in 1 blocks
==12345== total heap usage: 1 allocs, 0 frees
==12345==
==12345== LEAK SUMMARY:
==12345==    definitely lost: 16 bytes in 1 blocks
```

The tool is useful after the program compiles and the failure is runtime behavior.

EXPORT STRATEGY

ONE SOURCE, THREE OUTPUTS

render.sh

```
$ PATH="$PWD/venv/bin:$PATH" quarto render cm3.qmd --to revealjs
$ PATH="$PWD/venv/bin:$PATH" quarto render cm3.qmd --to beamer
$ PATH="$PWD/venv/bin:$PATH" quarto render cm3.qmd --to typst
```

Output	File	Use
RevealJS	cm3.html	live lecture with navigation, fragments, scroll, animation
Beamer	cm3-beamer.pdf	slide PDF close to the classroom deck

Output	File	Use
Typst	cm3- handout.pdf	printable reading handout from the same source

PRINT HANDOUTS FROM REVEALJS

RevealJS can be printed from the browser using `?print-pdf`.

```
handouts.sh
$ chromium --headless --no-sandbox \
  --print-to-pdf=cm3-reveal-print.pdf \
  "file://$PWD/cm3.html?print-pdf"

$ pdfjam cm3-reveal-print.pdf --nup 1x2 --paper a4paper --outfile cm3-2up.pdf
$ pdfjam cm3-reveal-print.pdf --nup 2x2 --paper a4paper --outfile cm3-4up.pdf
$ pdfjam cm3-reveal-print.pdf --nup 3x2 --landscape --paper a4paper --outfile c
```

These PDFs are for students who print. They should not depend on scrolling.

WHAT TO KEEP IN THE FINAL COURSE

Keep:

- code filenames and line highlights;
- vertical detail slides for optional explanations;
- one executed example per topic when it proves something;
- short command traces for terminal stories;
- Typst handouts for printable reading.

Avoid:

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- long scrollable slides as the only source of information;
- terminal styling that competes with syntax highlighting;
- animations that carry essential meaning with no PDF fallback.